



IILM UNIVERSITY

INTERNSHIP PRESENTATION

Rule-Based AI Order Support Bot

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Batch: 2CSE10

Organization Name: DecodeLabs AI

2. Internship Overview

Internship project focused on applying Python and introductory AI concepts.

Project: Rule-Based AI Order Support Bot.

Goal: automate common customer order-support queries.

Approach: predefined intents + pandas + orders.csv dataset.

3. Problem Statement

Customer-support teams receive repetitive order-related questions.

Manual responses can be slow and consume staff time.

A lightweight automated assistant can provide quick and consistent answers.

The prototype focuses on common order, tracking, history and spending queries.

4. Objectives

Build a simple rule-based AI chatbot in Python.

Read and process order information from a CSV dataset.

Recognize predefined user intents.

Support order ID and tracking lookup.

Show customer history and calculate total spending.

Handle greetings, exit commands and unknown queries.

5. Technologies Used

Python – core programming language.

Pandas – dataset loading, filtering and aggregation.

CSV – lightweight structured order data.

VS Code / Google Colab – development and execution environment.

Dictionary / if-else rules – intent detection.

6. Dataset Overview

Order ID, Date, Status and Fulfilment.

Sales Channel, shipping service level, SKU, Category and Size.

Quantity, Currency and Amount.

Shipping city, state, postal code and country.

Promotion IDs, B2B and fulfilled-by information.

7. System Workflow

1. Load orders.csv.
2. Accept user message.
3. Check greeting / exit commands.
4. Detect the most likely predefined intent.
5. Run the corresponding function.
6. Search or process the DataFrame.
7. Generate and display the response.
8. Continue until the user exits.

8. Main Features

Order ID lookup – retrieves matching order details.

Tracking lookup – returns available tracking/courier information.

Customer history – displays previous order records.

Total spend – sums relevant order amounts.

Coupon explanation – provides defined coupon information.

Fallback – handles unsupported questions.

9. Testing & Results

Greeting test → welcome response.

Valid order ID → matching record returned.

Tracking query → tracking information returned when available.

History query → customer orders displayed.

Spend query → total amount calculated.

Unknown query → fallback response.

Exit command → chatbot terminates correctly.

10. Advantages & Limitations

Advantages: simple, fast, explainable and easy to extend.

Does not require model training for the basic prototype.

Limitation: understands only predefined rules.

CSV is not ideal for large-scale production use.

No real-time courier API, voice interface or advanced NLP in the prototype.

11. Future Enhancements

NLP-based intent classification and fuzzy matching.

Multilingual support and voice interaction.

Web/mobile user interface.

Database integration and authentication.

Real-time courier API integration.

Conversation history, analytics and machine-learning upgrades.

12. Learning Outcomes

Improved Python programming and debugging skills.

Learned practical pandas DataFrame operations.

Understood intent detection and rule-based reasoning.

Learned how structured data can power an interactive application.

Improved documentation, testing and presentation skills.

13. Conclusion

The project demonstrates a practical AI-style customer-support solution.

Rule-based logic provides fast and understandable responses for supported queries.

The modular design creates a foundation for advanced NLP and production deployment.

The internship strengthened programming, data handling and problem-solving skills.